

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Captive breeding, habitat creation & management - Set 1

### Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding <b>(1)</b></li> <li>• Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) <b>(1)</b></li> </ul>                                                                                 | 2     |
| <b>01.2</b> | <i>Any 4 from the points below - 1 mark each, maximum 4.</i> <ul style="list-style-type: none"> <li>• Is the wild population threatened? <b>(1)</b></li> <li>• Is there a genetically diverse captive population? <b>(1)</b></li> <li>• Is in-situ conservation failing? <b>(1)</b></li> <li>• Is keeping a captive population realistic (habitat size, food, relationships, cost)? <b>(1)</b></li> <li>• Is release likely to be successful now or in future (suitable habitat, threats removed)? <b>(1)</b></li> </ul> | 4     |
| <b>01.3</b> | <i>2 marks for each point (partial credit for a partly correct point).</i> <ul style="list-style-type: none"> <li>• Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost <b>(2)</b></li> <li>• Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant <b>(2)</b></li> </ul>                                         | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.                                                                                                                                                                                                                                                                                                                                                                                                  |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Very small gene pool so individuals are closely related and inbreeding is likely <b>(1)</b></li> <li>• Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low <b>(1)</b></li> <li>• A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) <b>(1)</b></li> <li>• Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies <b>(1)</b></li> </ul>                     | 4     |
| <b>02.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided <b>(1)</b></li> <li>• Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life <b>(1)</b></li> <li>• Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) <b>(1)</b></li> <li>• Stress / disturbance <b>(1)</b></li> </ul>                                                                | 3     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection <b>(1)</b></li> <li>• Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) <b>(1)</b></li> <li>• So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females <b>(1)</b></li> </ul> | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>03.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) <b>(1)</b></li> <li>• Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds <b>(1)</b></li> <li>• Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups <b>(1)</b></li> </ul>                        | 3     |
| <b>03.2</b> | <p><i>Any 5 from the points below - 1 mark each, maximum 5.</i></p> <ul style="list-style-type: none"> <li>• Large enough suitable habitat for a viable population <b>(1)</b></li> <li>• Reliable food supply <b>(1)</b></li> <li>• Low predation risk / original threat removed <b>(1)</b></li> <li>• Suitable breeding sites <b>(1)</b></li> <li>• Water <b>(1)</b></li> <li>• Support of the local human population <b>(1)</b></li> <li>• Official / legal protection of habitat and species <b>(1)</b></li> </ul>                                                                                                                                                                        | 5     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>03.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed <b>(1)</b></li> <li>Identifies causes of failure so future releases can be improved / extra support given <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) <b>(1)</b></li> <li>The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino <b>(1)</b></li> <li>Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die <b>(1)</b></li> </ul>                                                                                                                                                                                                                                      | 3     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant <b>(1)</b></li> <li>Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees <b>(1)</b></li> <li>Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) <b>(1)</b></li> </ul>                                                                                                                                                                                                                            | 3     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild <b>(1)</b></li> <li>Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool <b>(1)</b></li> <li>Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident <b>(1)</b></li> <li>Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding <b>(1)</b></li> </ul>                                                                                                                                                  | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>05.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) <b>(1)</b></li> <li>Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) <b>(1)</b></li> <li>Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions <b>(1)</b></li> <li>Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) <b>(1)</b></li> </ul> | 4     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) <b>(1)</b></li> <li>Islands provide roosting / nesting sites safe from terrestrial predators (foxes) <b>(1)</b></li> <li>Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                      | 3     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Mobile species (birds, flying insects) colonise easily <b>(1)</b></li> <li>Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| <b>05.4</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| <b>06.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leafs <b>(2)</b></li> <li>Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development <b>(2)</b></li> <li>Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients <b>(2)</b></li> </ul> | 6     |
| <b>06.2</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>Food: wildflower grasslands for seed-eating birds; providing food species <b>(1)</b></li> <li>Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) <b>(1)</b></li> <li>Control of competitors: removing rhododendron so wildflowers get light <b>(1)</b></li> <li>Pollination: a range of flowering plants with different flowering times so pollinators have food all season <b>(1)</b></li> <li>Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens <b>(1)</b></li> </ul>                                                                                   | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| <b>07.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession <b>(1)</b></li> <li>Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>07.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out <b>(1)</b></li> <li>• Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles <b>(1)</b></li> <li>• Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes <b>(1)</b></li> <li>• Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species <b>(1)</b></li> </ul> | 4     |
| <b>07.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass <b>(1)</b></li> <li>• Choice depends on the vegetation to be controlled and the plants to be protected <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>07.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground <b>(1)</b></li> <li>• Heat stimulates dormant heather seeds to germinate so young heather re-establishes <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>08.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump <b>(1)</b></li> <li>• About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages <b>(1)</b></li> <li>• Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice <b>(1)</b></li> <li>• Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth <b>(1)</b></li> </ul>                                                                                                                                                                     | 4     |
| <b>08.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour <b>(1)</b></li> <li>• Fewer young trees eaten so trees and riverside vegetation recovered <b>(1)</b></li> <li>• More trees allowed beaver populations to increase; beaver dams increased wetland area <b>(1)</b></li> <li>• Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers <b>(1)</b></li> </ul>                                                                                                                                                                                                                   | 4     |
| <b>08.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species <b>(1)</b></li> <li>• Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                             | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |